

WASP-43b

R-2523

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140412 · created 2026-08-25T01:23:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456759.65593 +/- 0.00074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.097 +/- 0.044
Transit depth (Rp/Rs)^2	0.94 +/- 0.85 [%]
Orbital Inclination (inc)	77.68 +/- 2.93
Airmass coefficient 1 (a1)	0.833 +/- 0.019
Airmass coefficient 2 (a2)	0.14 +/- 0.18
Scatter in the residuals of the lightcurve fit is	2.22 %
Best Comparison Star	#2 - [557, 350]
Optimal Method	PSF photometry
Transit Duration (day)	0.019 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.097 ± 0.044 vs 0.1594 (deviation 1.4σ)
Duration fit vs published	0.019 d vs 0.0512 d (deviation 1.87σ)
Mid-time O–C	1.3 min
Depth SNR	2.2
Residual scatter	2.22 %

Light curve

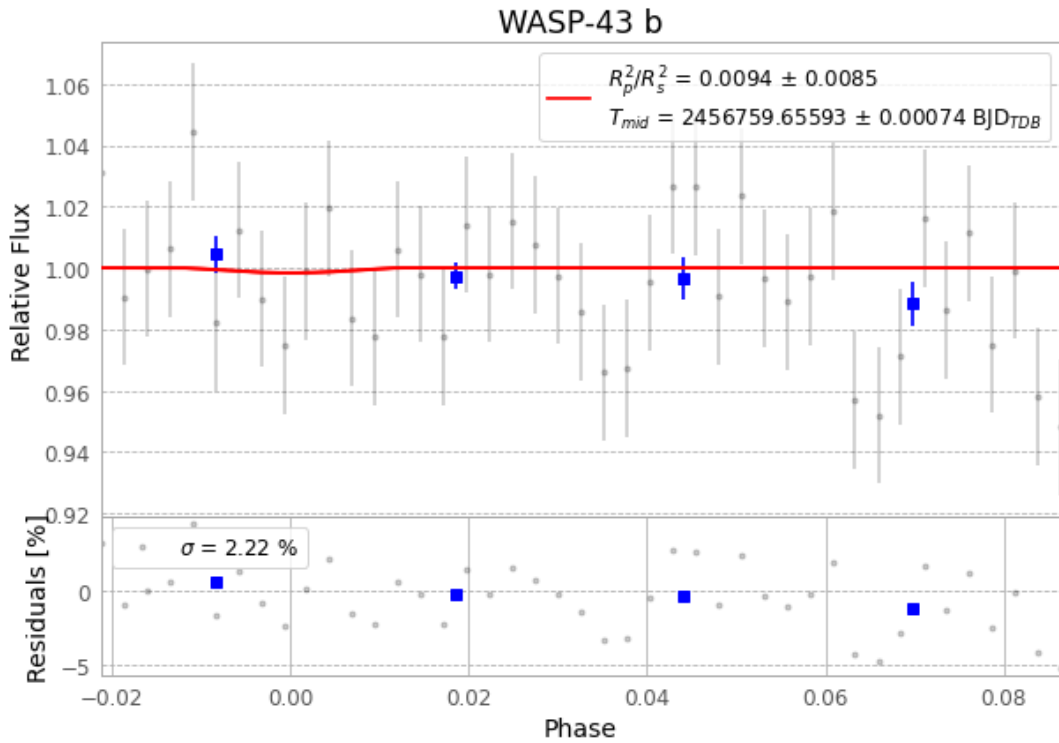


Figure 1 — FinalLightCurve_WASP-43 b_2014-04-11.png (SHA-256 d2f391e284a68213...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [176, 36], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2fca8298db2148fb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

43 FITS frames (28 MB), WASP-43140412031213.FITS → WASP-43140412051812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140412.log (47507b8e0273...), FinalLightCurve_WASP-43 b_2014-04-11.png (d2f391e284a6...), FinalLightCurve_WASP-43 b_2014-04-11.csv (c24c54a63660...)

Compute resources

Wall time 71.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 01:23Z.